

व्यावसायिक परीक्षण रिपोर्ट (प्रारंभिक)

COMMERCIAL TEST REPORT
(INITIAL)

संख्या/No: Machine-426/1264

माह/Month: March, 2021

Validity: 28.02.2026



FARMWORLD FWMT 210P POWER WEEDER



भारत सरकार

Government of India

कृषि एवं किसान कल्याण मंत्रालय

Ministry of Agriculture and Farmers Welfare

कृषि, सहकारिता एवं किसान कल्याण विभाग

Department of Agriculture, Cooperation and Farmers Welfare

दक्षिणी क्षेत्र कृषि मशीनरी प्रशिक्षण एवं परीक्षण संस्थान

Southern Region Farm Machinery Training and Testing Institute

ट्रैक्टर नगर, गार्लदिन्ने-515 731, जिला: अनंतपुर (आं. प्र.)

Tractor Nagar, Garladinne-515 731, District: Anantapur (A.P.)

[An ISO 9001:2015 CERTIFIED INSTITUTE]

Website: <http://srimtti.dacnet.nic.in/>

E-mail: fmti-sr@nic.in

Tele./FAX: 08551-286441

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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Applicant : M/s. Farmworld Agrotech pvt. Ltd,
SF 08, Shreemad Bhavan, Opp. Kanta
Stree Vikas Gruh, Dhebar Road,
Rajkot-360002, Gujarat, India

FARMWORLD FWMT 210P POWER WEEDER

Report No. : **Machine-426/1264**

Month: **March**

Year: **2021**



सत्यमेव जयते

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[An ISO 9001:2015 CERTIFIED INSTITUTE]

Telephone: 08551-286441

Website: <http://srfmtti.dacnet.nic.in/>

Fax: 08551-286441

E-mail: fmti-sr@nic.in

Machine-426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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Type of test : INITIAL COMMERCIAL TEST

Period of test : 08.02.2021 to 29.03.2021

Test Report No. : **Machine-426/1264**

Month / Year of release : March, 2021

- i) The results reported in this report are observed values and no corrections have been applied for atmospheric and site conditions.
- ii) The data given in this report pertain to the particular machine randomly selected by testing authority at applicant's site.
- iii) The results presented in this report do not in any way attribute to the durability of the machine.
- iv) This Test Report should not be reproduced in part or full without prior permission of the Director, Southern Region Farm Machinery Training & Testing Institute, Garladinne, Anantapur (A.P.).
- v) This is a report on Commercial Test of Power Weeder named "FARMWORLD FWMT 210P POWER WEEDER". **This report is valid up to 28.02.2026, Vide Ministry's O.M. No. 13-24/2018-M&T (I&P) dated 19.09.2018.**

SELECTED CONVERSIONS

S. No.	Units	Conversion Factor
1	Force	
	1 kgf	9.80665 N
		2.20462 lbf
2	Power	
	1 hp	1.01387 metric hp (Ps)
		745.7 W
	1 Ps	735.5 W
	1 kW	1.35962 Ps
3	Pressure	
	1 psi	6.895 kPa
	1 kgf/cm ²	98.067 kPa = 735.56 mm of Hg
	1 bar	100 kPa = 10 N/cm ²
	1 mm of Hg	1.3332 m-bar

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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Type of test : INITIAL COMMERCIAL TEST

Name of Machine : FARMWORLD FWMT 210P Power Weeder

Make : Farmworld

Model : FWMT 210P

Type : Self-propelled, Walk behind

Applicant : M/s. Farmworld Agrotech pvt. Ltd, SF 08,
Shreemad Bhavan, Opp. Kanta Stree Vikas
Gruh, Dhebar Road, Rajkot-360002, Gujarat,
India

TESTED BY

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Southern Region Farm Machinery Training and Testing Institute
Tractor Nagar, Garladinne-515 731, District: Anantapur (A.P.)
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Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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C O N T E N T S

Sl. No.	Chapters	Page No.
1	Scope of Test	6
2	Method of Selection	6
3	Test Code/Test Procedure	6
4	Specifications	7
5	Fuel & Lubricants	16
6	Running-in	16
7	Engine Performance Test	17
8	Amplitude of Mechanical Vibration	22
9	Noise Measurement	22
10	Air cleaner oil pull over test	23
11	Turning Ability	23
12	Hardness and Chemical Composition	24
13	Field Performance Test	25
14	Ease of Operation and Adjustments	26
15	Defects, Breakdowns and Repairs	26
16	Components / Assembly Inspection	26
17	Comments and Recommendations	29
	Annexure	31

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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1. SCOPE OF TEST

The scope of test was limited to check and assess the following:

- 1.1 Specification and other data furnished by the applicant.
- 1.2 Engine performance test
- 1.3 Amplitude of mechanical vibration
- 1.4 Noise measurement
- 1.5 Air cleaner oil pull over test
- 1.6 Turning ability
- 1.7 Hardness & chemical composition
- 1.8 Field performance
- 1.9 Wear analysis of rotor blades
- 1.10 Ease of operation and adjustments
- 1.11 Defects, breakdowns and repairs

2. METHOD OF SELECTION

The test sample was directly submitted by the applicant for test at this Institute. Hence the method of selection is not known.

3. TEST CODE/TEST PROCEDURE

There is no Indian standard/test code available for testing of self propelled power weeder as such. The guidelines, however, have been taken from the following:

- | | |
|-------------------------------------|---|
| IS 9935 : 2002
(Reaffirmed 2012) | : Power Tiller - Test code |
| IS 9980 : 1999
(Reaffirmed 2004) | : Guidelines for field performance and haulage tests of power tillers |
| IS 1976 : 1976
(Reaffirmed 2009) | : Specification for Rotary paddy weeder, manually operated |
| IS 6690 : 1981
(Reaffirmed 2012) | : Specification for Blades for Rotavator for Power Tillers |

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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4. SPECIFICATION

4.1 General:

Make (apa)	: FARMWORLD
Model	: FWMT 210P
Name of applicant	: M/s. Farmworld Agrotech pvt. Ltd, SF 08, Shreemad Bhavan, Opp. Kanta Stree Vikas Gruh, Dhebar Road, Rajkot- 360002, Gujarat, India
Name of machine	: FARMWORLD FWMT 210P Power Weeder
Type of machine	: Self propelled, Walk behind
Working width of machine, (mm)	: 750
Year of manufacture	: 2020
Serial no. of machine	: 202021356

4.2 Details of prime mover:

Make	: FARMWORLD
Model	: GK 210
Type	: Single cylinder, Four stroke, Air cooled, Inclined, Spark ignition engine
Year of manufacture	: 2020
Engine serial No.	: 202021356
Recommended high idle speed, rpm	: 3840 $\pm$ 50
Recommended low idle speed, rpm	: 1400 $\pm$ 100
Recommended rated speed, rpm	: 3600
Speed at maximum torque, rpm	: 2500
Engine speed for field test, rpm	: 3300
Max. power observed, kW (metric hp)	: 3.8 (5.2) at 3600 rpm

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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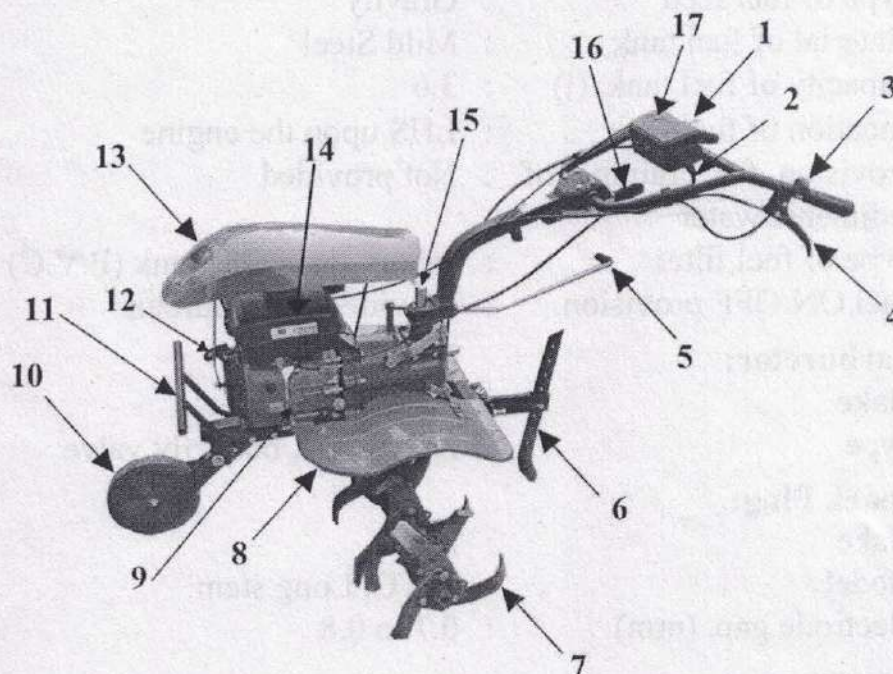


Fig.1. FARMWORLD FWMT 210P POWER WEEDER

KEYS:

- | | |
|-----------------------------|------------------------------------|
| 1. Accelerator | 10. Support wheel |
| 2. Reverse engaging lever | 11. Bumper |
| 3. Emergency stop switch | 12. Starting rope |
| 4. Main clutch | 13. Engine guard |
| 5. Main gear shifting lever | 14. Fuel tank |
| 6. Depth adjusting lever | 15. Handle rotation lever |
| 7. Rotary blade | 16. Handle height adjustment lever |
| 8. Rotary cover | 17. Tool box |
| 9. Engine | |

4.2.1 Cylinder and cylinder head:

Number	: One
Disposition	: Inclined
Bore/Stroke, (mm)	: 70/55
Capacity, cc	: 212
Compression ratio (apa)	: 8.7: 1
Type of cylinder liner	: Dry , Inbuilt
Valve clearance, (mm) (cold)	
Inlet	: 0.15 to 0.20
Exhaust	: 0.15 to 0.20

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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4.2.2 Fuel supply system:

Type of fuel feed	: Gravity
Material of fuel tank	: Mild Steel
Capacity of fuel tank, (l)	: 3.6
Location of fuel tank	: LHS upon the engine
Provision for draining of sediments/water	: Not provided
Type of fuel filter	: Inbuilt inside the tank (P.V.C)
Fuel ON/OFF provision	: Provided in carburetor

4.2.2.1 Carburetor:

Make	: S.P
Type	: Horizontal, butterfly valve

4.2.2.2 Spark Plug:

Make	: LG
Model	: F7TC, Long stem
Electrode gap, (mm)	: 0.7 to 0.8

4.2.3 Governor:

Make	: Not specified
Type	: Variable speed, Mechanical, centrifugal
Governed range of engine speed, (rpm)	: 1300 to 3890

4.2.4 Air-intake system:

4.2.4.1 Pre-cleaner : Not provided

4.2.4.2 Air-cleaner:

Make	: Not specified
Type	: Wet
Type of element	: Foam
Location of air-cleaner	: RHS front side & upon the engine
Capacity of air cleaner (l)	: 0.025
Recommended grade of oil & service schedule :	: After every 50h change oil and change foam element

4.2.5 Exhaust:

Type	: Up draft, square
Location	: RHS rear side upon engine
Provision against entry of rain water	: Provided
Provision for cover on exhaust	: Provided
Direction of exhaust emission	: Away from the operator

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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4.2.6 Lubrication system:

Type	: Splash
Type of lubricant recommended	: SAE 10W30
Oil capacity, (l)	: 0.450
Oil change period	: 50 hours
Max. permissible oil pressure (Kgf/cm ²)	: Not specified
Relief valve sitting (Kgf/cm ²)	: Not specified

4.2.7 Cooling system:

Type	: Air cooled
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4.2.7.1 Flywheel & Fins details:

Material	: HDPE
No. of fins (Ribs)	: 24
Length x width x thickness of fins, (mm)	: 58x26.56x3.08 (t)
Dia. of flywheel (mm)	: 171.5
Width of fly wheel along with inbuilt fins	: 72.12

4.2.8 Starting system:

Type	: Manual, Recoil
Ignition system	: Spark ignition
Aid for cold starting	: Choke provided
Any other provision for easy starting	: None

4.3 Power transmission system:

4.3.1 Clutch

Make	: Not specified
Type of clutch	: Wet, friction multi plate
Dia. of clutch plate, (mm)	: 110.90
No. of friction plates	: 05
No. of pressure plates	: 06
No. of coil springs	: 04
Material (apa)	: Not specified
Thickness of the clutch plate, (mm)	: 2.63
Location of clutch	: Inside the gear box

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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4.3.2 Primary (Main) gear

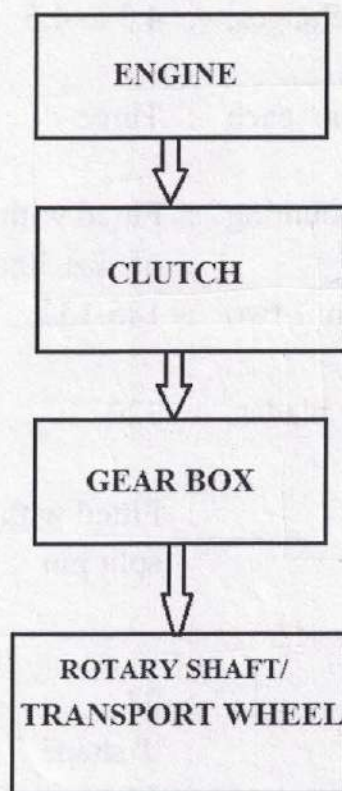
box:

Make	: Farmworld
Model	: Not specified
Type	: Sliding mesh
No. of speeds	: 03 (2 forward and 1 reverse)
Grade of oil	: SAE 90
Oil capacity, (l)	: 1.3
Oil changing period	: 50 hours
Reduction ratio :	
Forward	
1 st gear	: 2.438:1
2 nd gear	: 1.500:1
Reverse	: 3.352:1
No. and size of bearing	: Ball bearings 6203PG-02Nos.& 6007-01No.; Thrust bearing -01 No. ; Tapper roller bearing 30203 -01 Nos.

4.3.3 Secondary reduction box for rotor:

Type	: Geary type (Bevel & pinion)	
	I st Pair	II nd Pair
No. of teeth on drive gear	: 8	8
No. of teeth driven gear	: 35	30
Reduction ratio	: 4.375:1	3.75:1
Overall reduction ratio from engine crankshaft to rotor shaft		
	Forward	
	1 st	: 39.998:1
	2 nd	: 24.609:1
	Reverse	: 54.998:1
No. of bearing	: Tapper roller bearing 30205, 30204 & 30305-03Nos. ; Ball bearing 6305PG-01No.	

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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**Fig. 2 Schematic power flow diagram
(Farmworld FWMT 210P Power Weeder)**

4.4 Rotary unit

Make : Not specified

4.4.1 Rotary cover :

Material : Mild steel

Size, (mm) : 490x280

Thickness of sheet, (mm) : 1.0

Method of fixing : Fitted upon fabricated MS Angle with the help of 4+4 Nos. bolt and nuts of size 16xØ5.8

4.4.2 Rotary shaft:

Material (apa) : Not specified

Type of rotor axle : Hexagonal

Size of shaft, (mm) length of shaft, mm : 295xØ22.80

No. of flanges : 4+4

Type of flanges : Rectangular

Size of flanges, (mm) : 100x90

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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Thickness of flanges, : 4.3 to 4.5
(mm)

No. of blades on each : Three
flange

Method of mounting : Fitted with the help of 2 Nos. of bolt and nuts
blades on flanges of size 30xØ9.75

Distance between two : 125-132
flanges, (mm)

Dia. of rotor with blades, : 320
(mm)

Method of fixing : Fitted with help of cotter pin and locked with
split pin

4.4.3 Rotary blades:

Numbers : 24

Type : 'J' shape

Material (apa) : 65 Mn

Thickness, (mm) : 4.95

Width of beveled edge, : 7.00 to 10.23
(mm)

No. & size holes on each : Two , Ø 10.84
blade for fixing it to the
flanges, (mm)

Arrangement of blades on : Alternately on LHS & RHS
the axle

4.5 Details of transport wheel:

Make of tyre : RH

Number & Type : Two, pneumatic traction

Size & Ply rating : 4.00 – 8 & 4 PR

Max. loading cap. : 265Kg at 280kPa

Method of fixing : Fitted with help of cotter pin and locked with
split pin

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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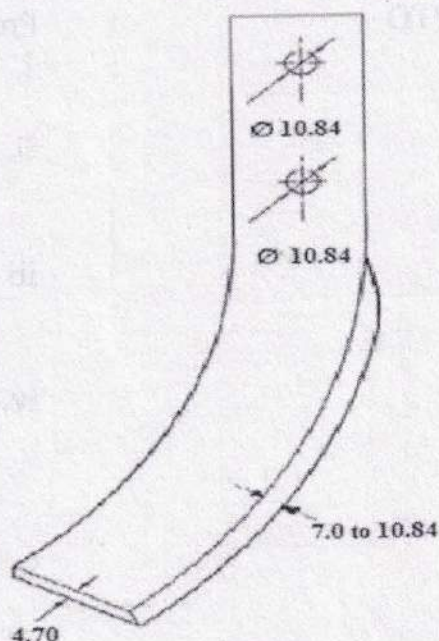


Fig. 3 Rotary blade
(All dimensions in mm)

4.6

Depth control mechanism:

Type	:	MS flat
Material	:	Mild Steel
No. of flat (s)	:	One
Size of flat, (mm)	:	
Curved length	:	480
Width	:	38.60
Thickness	:	11.60
Provision for depth adjustment	:	8 no. of holes of size Ø 10.80 mm are provided at distance of 40mm

4.7

Steering handle bar:

No. of steering handle	:	Two
Material	:	Mild steel
Dia. of pipe (OD), (mm)	:	25.0
No. of hand grips	:	Two
Material of grip	:	Rubber
Length of grip, (mm)	:	122
Dia. of grip, (mm)	:	32.59/38.98
Height of handle bar from ground level, (mm)	:	750 to 1380
Provision for handle height adjustment	:	Provided in 4 positions
Provision for handle rotation	:	Provided

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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- 4.8 Provision for PTO : Provided
- Controls :
- Clutch lever is provided on LHS handle bar.
 - Accelerator lever is provided on RHS of handle bar.
 - Emergency stop switch is provided on LHS of handlebar.
 - Reverse gear engaging and disengaging lever is provided on RHS of handle bar.
 - Main gear shifting lever is provided at below RHS of handle bar.
 - Handle height adjustment lever is provided at centre handle bar.
 - Handle rotation lever is provided on center of handle bar.
 - Tool box is provided at centre of handle bar.

4.9 Marking/Labeling:

A metallic labeling plate is riveted on Tool bar / handle bar base with the following information:

Honour Your Trust Mini Tiller		
Model	:	FWMT210P
Engine	:	4 stroke, Petrol engine
Rated power	:	4.4kW/3600 rpm
Displacement	:	212
Tilling depth	:	100-250mm
Tilling width	:	750-1050mm
Sr. No.	:	202021356
Year	:	2020
Farmworld Agro Private Limited Rajkot, Gujarat, India www.farmworld.in		

Machine-426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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- 4.10 Overall dimensions, (mm):**
Length : 1480
Width :
- With rotary : 1170
- With transport wheels : 640
Height : 1380
- 4.11 Mass, (kg)** :
With rotary axle : 69.5
With transport wheel : 70.0
- 4.12 Colour of machine:**
Engine : Silver
Main Drive Cover, Transmission : Pale green
and Rotary cover & Flywheel cover
Silencer, Air cleaner, Handle bar and : Black
chassis

5. FUEL & LUBRICANTS

5.1 Fuel : Petrol

5.2 Lubricants

Sl. No.	Particulars	As recommended by the manufacturer	As used during the test
1	Engine	SAE 10W30	As per Applicant's Recommendation
2	Air cleaner	SAE 10W30	
3	Gear box	SAE 90	

6. RUNNING-IN

The power weeder was Run-in for 1.00 hours before the actual test. All the fasteners were checked and tightened thereafter.

7. ENGINE PERFORMANCE TEST

Date(s) of test : 15.03.2021 and 17.03.2021
Power Weeder run at the Institute : 34.5
before engine performance test (h)
Dynamometer : EC-30, Eddy Current

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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7.1 The results of Engine performance tests are tabulated in **Table-1** and graphically represented in **Fig. 4, 5 and 6**.

Table – 1

Power, (kW)	Crank shaft speed, (rpm)	Fuel consumption			Specific energy, (kWh/l)
		(l/h)	(kg/h)	Specific, (kg/kWh)	
1	2	3	4	5	6
Maximum power – 2 hours test:					
3.8	3600	2.24	1.67	0.440	1.7
3.7	3601	2.13	1.59	0.429	1.7*
Power at rated engine speed (3600 rpm):					
3.8	3600	2.24	1.67	0.440	1.7
3.7	3601	2.13	1.59	0.429	1.7*
Maximum torque:					
3.8	3600	2.24	1.67	0.440	1.7
3.7	3601	2.13	1.59	0.429	1.7*
Part loads:					
i) At torque corresponding to maximum power:					
3.8	3600	2.24	1.67	0.440	1.7
ii) 85% of the torque obtained in (i)					
3.38	3691	1.77	1.32	0.391	1.9
iii) 75% of the torque defined in (ii)					
2.6	3749	1.45	1.08	0.420	1.8
iv) 50% of the torque defined in (ii)					
1.7	3785	1.04	0.78	0.448	1.7
v) 25% of the torque defined in (ii)					
0.9	3819	0.90	0.67	0.764	1.0
vi) Unloaded					
--	3852	0.72	0.54	--	--

*High ambient conditions.

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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		<u>Natural ambient</u>	<u>High ambient</u>
- No load max. engine speed, (rpm)	:	3852	3858
- Crankshaft torque at max. Power, (Nm)	:	10.0	9.1
- Max. crankshaft torque, (Nm)	:	10.0	9.1
- Engine speed at max. crankshaft torque, (rpm)	:	3600	3600
- Back-up torque, (%)	:	Nil	---

Range of atmospheric conditions:

- Temperature (°C)	:	30 to 32	41 to 44
- Pressure, (kPa)	:	97.2	97.1 to 97.2
- Relative humidity, (%)	:	20 to 26	20 to 26

Max. Temperatures (° C):

- Engine oil	:	82	88
- Fuel	:	32	45
- Air intake	:	32	45
- Exhaust gas	:	342	448

Pressure at max. power:

- Intake air, (kPa)	:	---	---
- Exhaust gas, (kPa)	:	3.5	4.6

Consumptions:

- Lub oil, (g/kWh)	:	---	3.06
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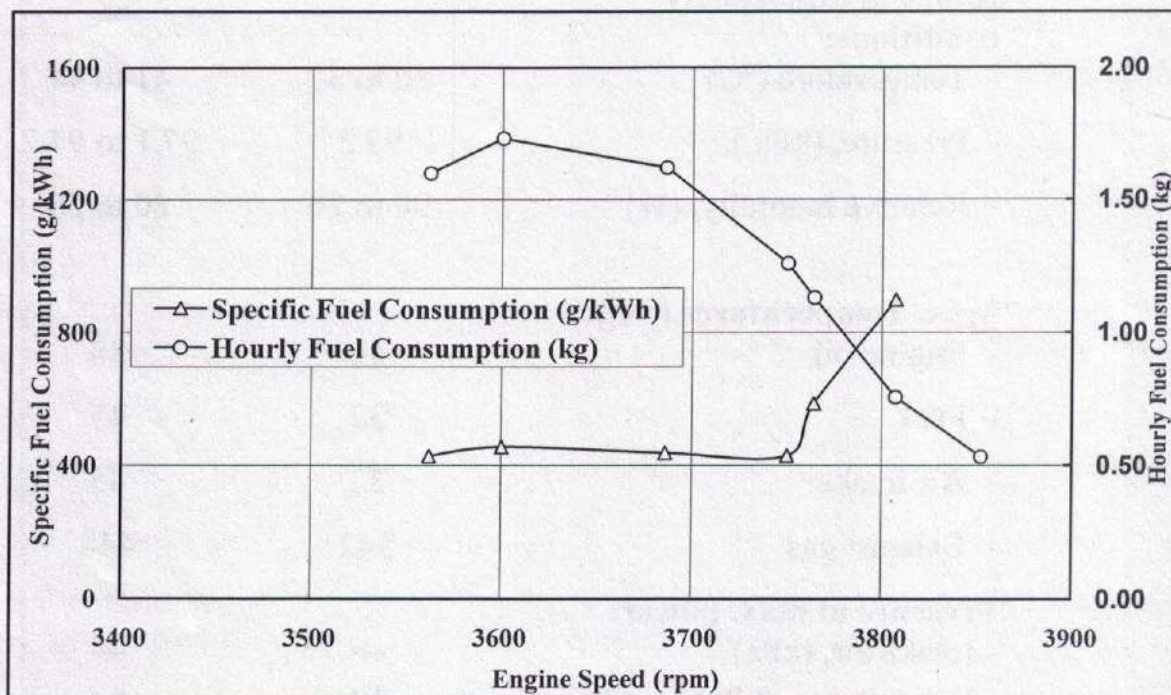
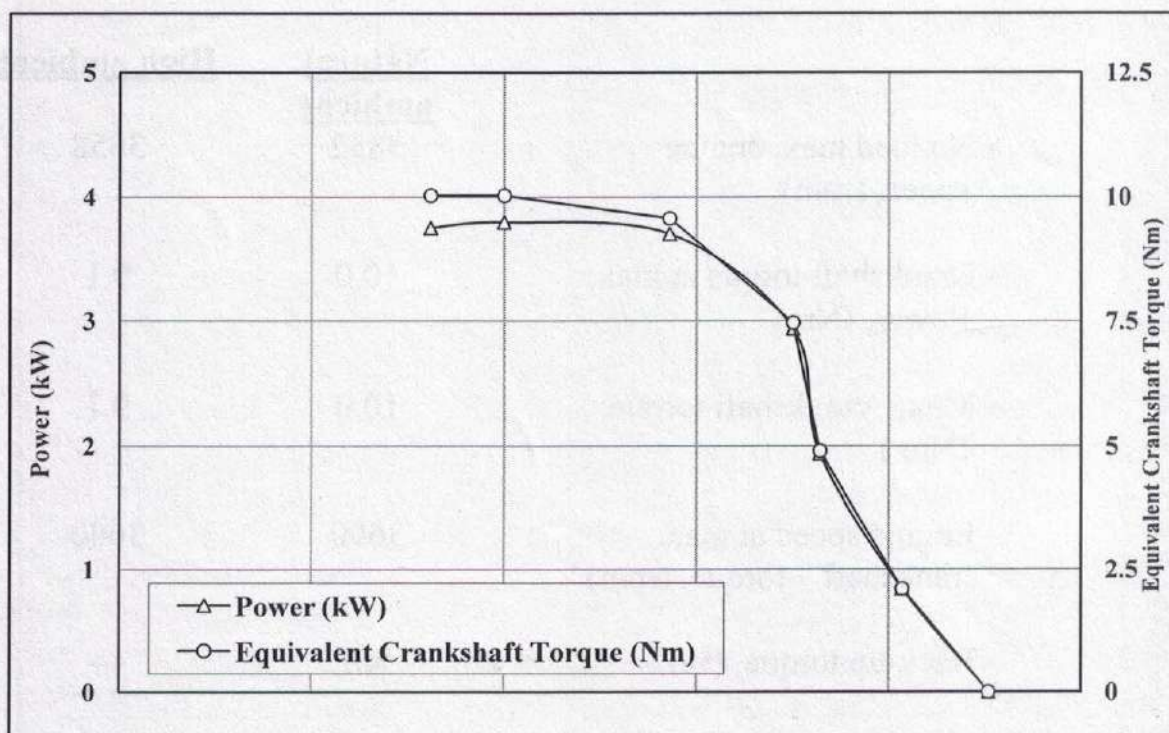


Fig. 4 ENGINE PERFORMANCE TEST (NATURAL AMBIENT)

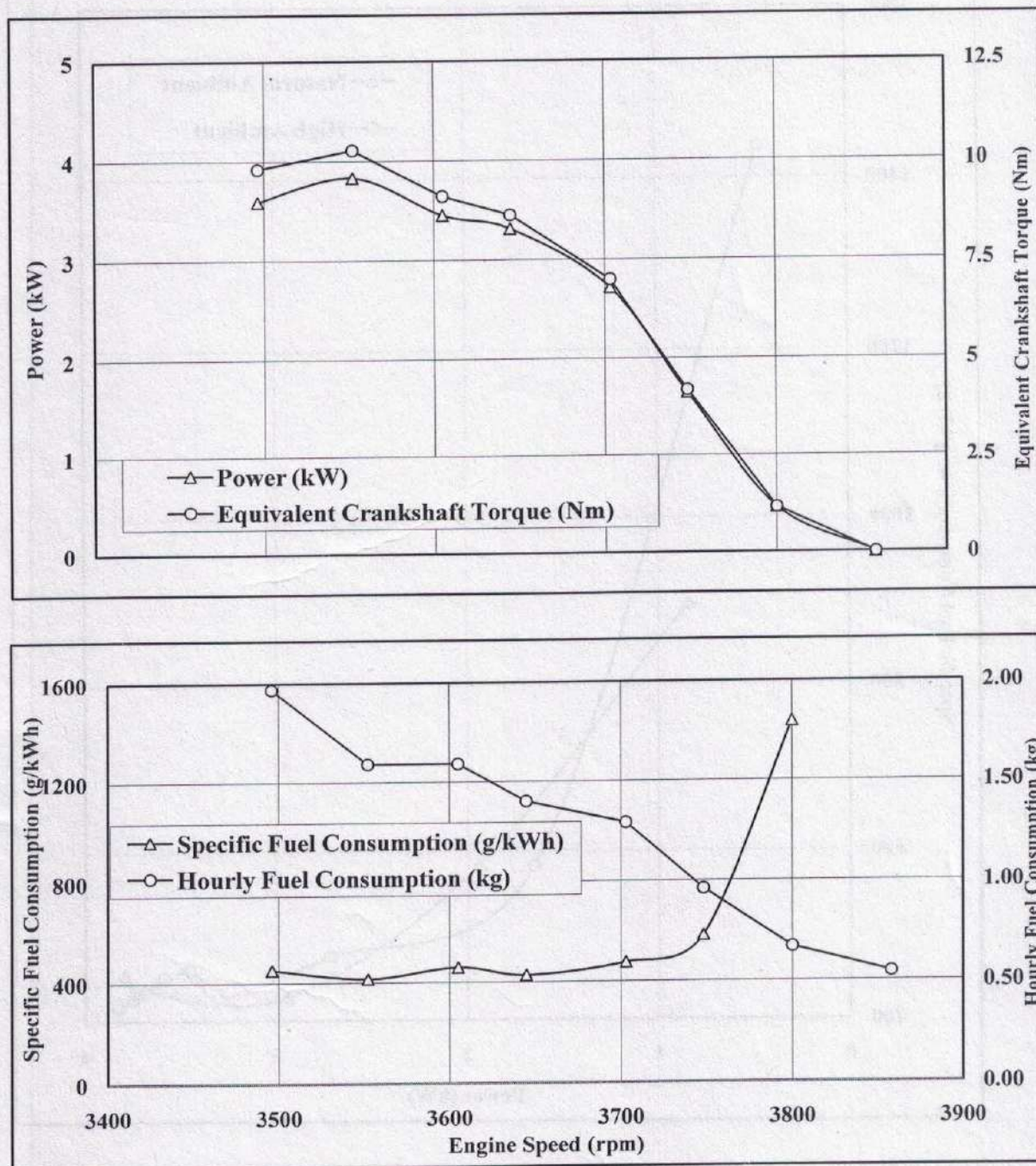


Fig. 5 ENGINE PERFORMANCE TEST (HIGH AMBIENT)

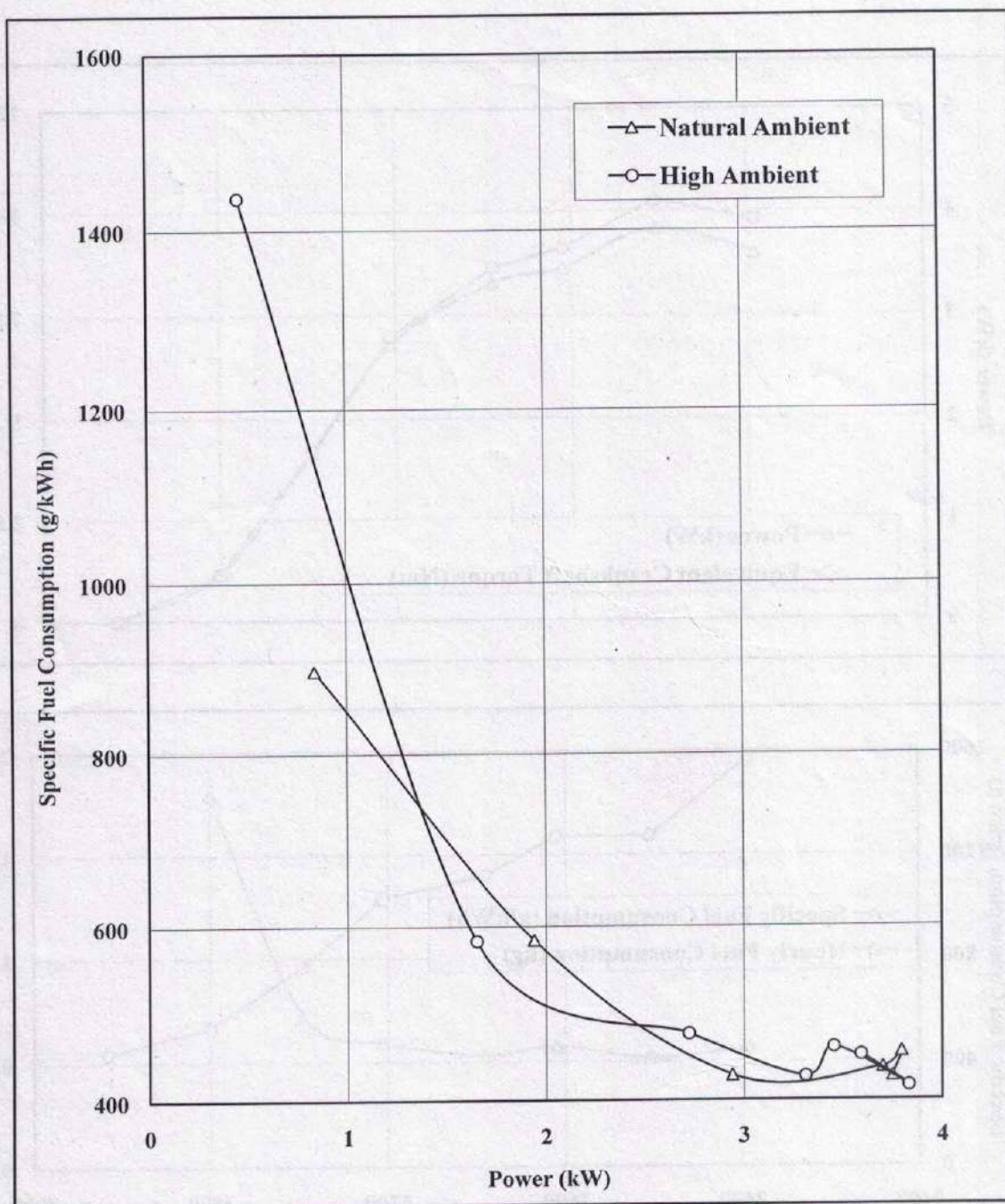


Fig. 6 ENGINE PERFORMANCE CHARACTERISTICS

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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8. AMPLITUDE OF MECHANICAL VIBRATION

Date of test : 22.02.2021
 Type of test surface : Concrete
 Type of instrument used : VM-30
 Test condition : At rated engine speed without load and all components are in working position.

Sl. No.	Location		X (μ)	Y (μ)
1.	Steering handle	Left	110*	140*
		Right	100	110*
2.	Main clutch lever		120*	130*
3.	Fuel tank cap		170*	198*
4.	Fuel tank		340*	370*
5.	Gear shifting lever		50	60
6.	Rotary shield cover		330*	350*
7.	Tool bar		80	90
8.	Air cleaner		180*	220*
9.	Engine mounting base		260*	290*
10.	Silencer		430*	480*
11.	Accelerator lever		110*	130*
12.	Handle bar		110*	120*

* The amplitude of Mechanical Vibration is on higher side

9. NOISE MEASUREMENT

Date of test : 22.02.2021
 Type of sound level meter : B&K, type 2239
 Type of test track : Loose Soil
 Height of microphone, (m) : 1.2
 Test condition : At rated engine speed

Atmospheric conditions

Temperature, °C : 32
 Pressure, kPa : 97.2
 Relative humidity, % : 53.4
 Background noise level, dB(A) : 50
 Wind velocity, m/s : 1.5
 Observed noise level, dB(A)
 - By stander's position : 72
 - Operator's ear level : 88

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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10. AIR CLEANER OIL PULLOVER TEST

Date of test : 09.02.2021
 Range of atmospheric conditions:
 Temperature, (°C) : 31 to 34
 Pressure, (kPa) : 97.1 to 97.3
 Relative humidity, (%) : 26 to 37
 Mass of oil before test, (g) : 22.8

Sl. No.	Position of power weeder	Loss of oil (g)	Oil pullover (%)
i)	Power Weeder parked on level ground	0.2	0.88
ii)	Power Weeder tilted 15° laterally on RHS	Nil	Nil
iii)	Power Weeder tilted 15° laterally on LHS	0.2	0.88
iv)	Power Weeder tilted 15° longitudinally with front end up	0.3	1.31
v)	Power Weeder tilted 15° longitudinally with rear end up	Nil	Nil

11. TURNING ABILITY

Date of test : 10.02.2021
 Type of test track : Concrete

The minimum turning diameter and the minimum clearance diameter of the machine was measured and recorded as under:

Characteristics	Minimum turning diameter, m	Minimum clearance diameter, m
LHS	0.95	2.23
RHS	0.95	2.37

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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12. HARDNESS AND CHEMICAL COMPOSITION

12.1 Hardness of rotor blades :

The surface hardness of blade was recorded as under: -

Sl. No.	As per IS 6690:1981 (Reaffirmed 2012)	As observed (HRC)	Remarks
1	56 ±3 HRC	54.0	Conforms
2		54.6	
3		55.0	
4		56.0	
5		57.0	

12.2 Chemical composition of rotor blades :

The material of rotary blade was got analyzed from Geological and Metallurgical Laboratories, Bangalore for chemical composition. The results of chemical analysis test results vide test report No. GML 5946-5/2020-2021, dated 17.02.2021 is as under:

Constituents	As per IS 6690:1981 (Reaffirmed 2012)		Composition as observed (% of weight)	Remarks
	Carbon Steel	Silicon Manganese Steel		
Carbon (C)	0.70 -0.85	0.50-0.60	0.578	Conforms
Silicon (Si)	0.10 -0.40	1.50-2.00	1.922	Conforms
Manganese (Mn)	0.50 -1.0	0.50-1.00	0.744	Conforms
Sulphur (S)	0.05(max)	0.05(max)	0.006	Conforms
Phosphorous (P)	0.05(max)	0.05(max)	0.028	Conforms
Boron (B)	---	---	<0.001	---

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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13. FIELD PERFORMANCE TEST

The field tests were conducted for 26.0 hours for testing the said Power Weeder. During the field tests no load engine speed adjusted to 3498 to 3518 rpm. In all, five tests were conducted in medium soil. The detailed test results are represented in the Annexure and summarized in the ensuing table:

Sl.No.	Parameters	Range
1	Type of soil	Medium
2	Av. soil moisture (%)	7.2 to 12.8
3	Av. bulk density of soil (g/cc)	1.55 to 1.87
4	Av. speed of operation (kmph)	1.26 to 1.38
5	Av. depth of cut (cm)	6.1 to 6.8
6	Av. width of cut (cm)	110 to 113
7	Av. area covered (ha/h)	0.094 to 0.108
8	Av. time required for one ha (h)	9.26 to 10.64
9	Av. field efficiency (%)	67.62 to 69.54
10	Av. weeding efficiency (%)	58.01 to 65.55
11	Av. fuel consumption	
	- l/h	1.20 to 1.50
	- l/ha	11.48 to 15.15

13.1 Rate of work

- Av. rate of work was recorded as 0.094 to 0.108 ha/h and the speed of operation vary from 1.26 to 1.38 kmph.
- Av. time required to cover one hectare was recorded as 9.26 to 10.64h/ha.

13.2 Quality of work

- Av. depth of cut was recorded as 6.1 to 6.8 cm.
- Av. working width was observed as 110 to 113 cm.
- Av. field efficiency was found as 67.62 to 69.54 %.
- Av. weeding efficiency was found as 58.01 to 65.55 %.

13.3 Adequacy of power of prime mover

The power of prime mover was found adequate.

SOUTHERN REGION FARM MACHINERY TRAINING & TESTING INSTITUTE, ANANTAPUR, (A.P.)	Page 25 of 31
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Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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13.4 Wear Analysis of rotor blades

Sl. No.	Initial mass(g)	Final mass (g)	Loss of mass (g)	Percentage of wear	
				After 26 h	Per hour
1	310.4	301.5	8.90	2.87	0.11
2	336.2	326.6	9.60	2.86	0.11
3	328.3	316.3	12.00	3.66	0.14
4	338.2	314.3	23.90	7.07	0.27
5	325.3	317.0	8.30	2.55	0.09
6	341.1	333.0	8.10	2.37	0.09
7	325.9	323.4	2.50	0.77	0.03
8	327.4	315.2	12.20	3.73	0.14

14. EASE OF OPERATION & ADJUSTMENTS

No noticeable difficulty was observed during the operation & adjustment of the machine.

15. DEFECTS, BREAKDOWNS AND REPAIRS

No defects & breakdowns was observed during the entire test.

16. COMPONENTS / ASSEMBLY INSPECTION

The Power Weeder was dismantled after 44.47 hours of operation at this Institute.

16.1 Engine:

Cylinder bore:

Cyl in d er N o.	Cylinder bore dia, (mm)						Max. permissible wear limit, (mm)
	Top position		Middle position		Bottom position		
	Thrust Side	Non- thrust Side	Thrust side	Non- thrust Side	Thrust Side	Non- thrust side	
1.	70.0	70.0	70.0	70.01	70.02	70.01	70.40

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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Piston:

Pist on No.	Piston dia (mm)				Clearance between piston & cylinder liner at the skirt of the piston, (mm)	Maximum permissible clearance limit, (mm)
	Top(above top compression ring)		At skirt			
	Thrust side	Non- thrust Side	Thrust side	Non- thrust side		
1.	69.45	69.46	69.90	***	0.12	0.90

*Not measured due to piston design constraints

Ring end gap:

Rings	Ring end gap, (mm)			Max. permissible end gap limit, (mm)
	Top	Middle	Bottom	
1 st comp. Ring	0.22	0.23	0.24	0.30
2 nd comp. Ring	0.22	0.23	0.25	
Oil ring	0.24	0.25	0.27	

Ring side clearance:

Rings	Ring side clearance, (mm)	Max. permissible clearance limit, (mm)
1 st Compression ring	0.04	0.40
2 nd Compression ring	0.04	
Oil ring	0.07	

Main and big end bearings:

Main bearings:

Bearing No.	Diametrical clearance, (mm)	Crankshaft end float, (mm)	Max. permissible clearance limit,(mm)	
			Diametrical clearance	Crankshaft end float
1.	Ball bearing	0.12	NA	0.20
2.	Ball bearing			

Big end bearings:

Bearing No.	Clearance, (mm)		Max. permissible clearance limit,(mm)	
	Diametrical	Axial	Diametrical	Axial
1.	0.09	0.45	0.025	0.15

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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	Valve, guides and timing gears:	Observation
	Any marked sign of overheating of valves	: None
	Pitting of seat/faces of valves	: None
	Any visual damage to the teeth of timing gears	: None
	Clearance between valve guide and valve stem (mm)	
	- Intake valve	: 0.10
	- Exhaust valve	: 0.11
16.2	Clutch:	
	Any marked wear in clutch friction plate	: None
	Condition of clutch release bearing.	: Normal
	Condition of pilot bearing	: Normal
	Condition of pressure plate	: Normal
	Presence of oil and water in clutch housing	: None
16.3	Transmission gears:	
	All the Gears and sprockets of the transmission system were found in normal condition.	
16.4	Rotary drive unit:	
	The rotary drive unit was dismantled and all the components were found in normal condition.	

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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17. COMMENTS & RECOMMENDATIONS

17.1 Engine performance:

- 17.1.1 The maximum power and rated power were observed as 3.8 kW under natural ambient condition against the declared value of 4.4 kW. This should be looked into for corrective action.
- 17.1.2 Specific fuel consumption of engine corresponding to maximum power was recorded as 440 g/kWh against the declared value of 380 g/kWh. This should be looked into for corrective action.
- 17.1.3 Max. Torque was observed as 10.0 Nm against the declared value of 11.0 Nm.
- 17.1.4 Back up torque of engine was observed as Nil against the declared value of 4.0 %. This should be looked into for corrective action.
- 17.1.5 **During varying speed test of engine at both natural & high ambient conditions after attaining max. torque at 10.0 Nm & 9.1 Nm respectively, while further loading, black smoke and sudden drop of engine rpm was observed. The Governor hunting was also observed after rated engine speed.**
- 17.2 Max. Noise at operator's ear level was observed as 88 DB (A).
- 17.3 The amplitude of mechanical vibration on most of the assemblies of the Power weeder exceeds the limit of 100 micron. Therefore necessary action to reduce vibration should be taken.
- 17.4 The hardness of rotary blades conforms to the requirement of IS 6690:1981 (Reaffirmed 2012).
- 17.5 Chemical composition of the rotary blade conforms to the requirement of IS 6690:1981 (Reaffirmed-2012).
- 17.6 The provided labeling plate should be as per the requirement IS.

Machine- 426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
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17.7 Technical literature:

Operator's & service manual, multifunctional manual and spare parts catalogue are required to provide in separate booklet with adequate related information in separate booklet. It is however, recommended that same may be revised and brought out in Hindi & other regional languages also as per IS 8132:1999 (Reaffirmed 2004) for the sake of user & technical personnel.

TESTING AUTHORITY:

B.N. DIXIT AGRICULTURAL ENGINEER	
P.C. MESHRAM SENIOR AGRICUTRAL ENGINEER	
Dr. P.P. RAO DIRECTOR	

18. APPLICANT 'S COMMENTS

Para No	Our reference	Applicant's Comments
18.1	17.1.1	We will work on comments for better performance of engine, in support with our R &D and Production team. We will make all possible necessary changes our product future.
18.2	17.1.2	We will work on comments for better result of fuel consumption, In Support with our R &D and Production team. We will make all possible necessary changes our product future.
18.3	17.1.4	We will work on comments for better result of back up torque of engine, In support with our R &D and Production team. We will make all possible necessary changes in our product in future.
18.4	17.1.5	We will work on comments for better results in with support with R&D and Production team. We will make all possible necessary changes in our product in future.
18.5	17.2	In future we will give positive efforts to decrease further and maintain standards which are specified by ILO. We will make all possible necessary changes our product in future.

SOUTHERN REGION FARM MACHINERY TRAINING & TESTING INSTITUTE, ANANTAPUR, (A.P.)	Page 30 of 31
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Machine-426/1264	FARMWORLD FWMT 210P POWER WEEDER	COMMERCIAL (ICT)
18.6	17.3	We will work on it and take necessary steps, in support with our R & D and Production Team. We will provide anti vibration rubber grips at handle bars for operator comfort, safety and health. We will also look into adding vibration damping gaskets at all joints wherever possible. We make all possible necessary changes in our product in future.
18.7	17.6	We will provide as per the IS. We will make all possible necessary changes in our product in future.
18.8	17.7	Yes, We provided English & Hindi Language Operational manual, Service manual and Part catalogue of power weeder & Engine during the test. We will also look into Multiple Language Literature as per IS 8132:1999(Reaffirmed 2004)

ANNEXURE

FIELD PERFORMANCE RESULTS

Place of test: Institute farm fields, Anantapur, A.P.

Sl. No.	Parameters	I	II	III	IV	V
1	Date of test	11.02.21	12.02.21	13.02.21	14.02.21	15.02.21
2	Net test duration (h)	5	5	7	5	4
3	Furrow length (m)	93	85	90	88	90
4	Type of soil	Medium				
5	Bulk density (g/cc)	1.55	1.84	1.87	1.82	1.80
6	Soil moisture (%)	7.2	12.3	12.8	12.1	11.8
7	Previous treatment	NIL				
8	Forward speed, (kmph)	1.26	1.33	1.34	1.38	1.38
9	Av. depth (cm)	6.4	6.8	6.8	6.1	6.3
10	Av. width (cm)	110	110	113	110	113
11	Area covered (ha/h)	0.094	0.099	0.105	0.105	0.108
12	Time required for one ha (h)	10.64	10.10	9.52	9.52	9.26
13	Field efficiency (%)	67.62	67.81	69.54	69.08	69.23
14	Av. height of weeds, (cm)	13.4	15.2	12.6	16.6	16
15	Av. number of weeds per m ² (Before operation)	55.67	62.33	79.33	74.33	60.33
16	Av. number of weeds per m ² (After operation)	20.33	22.33	27.33	27.67	25.33
17	Weeding efficiency (%)	63.48	64.17	65.55	62.77	58.01
18	Fuel consumption					
	- l/h	1.20	1.50	1.25	1.30	1.24
	- l/ha	12.77	15.15	11.90	12.38	11.48



भारतसरकार

Government of India

कृषि एवं किसान कल्याण मंत्रालय

Ministry of Agriculture and Farmers Welfare

कृषि एवं किसान कल्याण विभाग

Department of Agriculture and Farmers Welfare

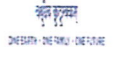
दक्षिणी क्षेत्र कृषि यंत्र प्रशिक्षण एवं परीक्षण संस्थान,

Southern Region Farm Machinery Training & Testing Institute

ट्रैक्टर नगर, गार्लदिन्ने, जि. अनंतपुर (आं. प्र.)

Tractor Nagar, Garladinne, District: Anantapur (A.P.)- 515731

An I.S.O. 9001- 2015 Certified Institute

Web.: <http://srfmtti.dacnet.nic.in>E-mail: fmti-sr@nic.in

File No. 16-99/2023-24-Test

Dated: 30.07.2024

To

M/s. FarmworldAgrotech Pvt. Ltd.,
SF 08, Shreemad Bhavan,
Opp. KantaStree Vikas Gruh, Dhebar Road,
Rajkot – 360 002, Gujarat, India.

Subject:-Extension of validity ICT tested Power Weeder– regarding.

Sir,

With reference to your email dated 27.12.2023 regarding the subject cited above, it is to state that your tested samples come under the category of **Self Propelled machinery** that have seven years of validity for ICT test as per Ministry's letter No.13-22/2020 -M&T (I&P) dated 12th December, 2023.

Sl. No	Name of the Machine	Make & Model	Type of Test	Test Report Number	Report Release Date	Validity of Test Report	Extended date of Validity
1	Power Weeder	FARMWORLD, FWMT420DE	ICT	Machine-422/1256	March, 2021	28.02.2026	28.02.2028
2	Power Weeder	FARMWORLD FWMT 210P	ICT	Machine-426/1264	March, 2021	28.02.2026	28.02.2028

(Pramod Yadav)
Agricultural Engineer
For Director

Copy to :

1. Applicant copy.
2. Correspondence file.

कृषि अभियंता /Agricultural Engineer
भारत सरकार/Government of India
द क्षे.का.म.प्र.एव.परी.संस्थान/S.R.F.M.T.&T.I
गार्लदिन्ने/GARLADINNE-515731
अनंतपुर जिला.(आं.प्र.)/ANANTAPUR DIST. (A.P.)